

A1:INTERACTIVE DATA DASHBOARD

- data integrated from two sources: one of the provided data sets and one additional external, public data set

To create the dashboard, I have integrated data from

1. The telecommunication industry Churn dataset (Churn_clean.csv). The dataset has 10000 rows and 50 features
2. External data from Kaggle USA States and regions (states.csv). The dataset has 51 rows and 4 features.

The screenshot shows the Tableau Desktop interface. The top menu bar includes File, Data, Server, Window, and Help. The sidebar on the left has a 'Connections' section with 'churn_clean' (Text file) and a 'Files' section with 'churn_clean.csv' and 'states.csv'. The main workspace displays a relationship diagram with 'churn_clean.csv' and 'states.csv' connected by a line. Below the diagram, a dropdown menu shows 'churn_clean.csv' and 'states.csv'. The 'Performance Options' section is expanded, showing 'Cardinality' set to 'Many' and 'Referential Integrity' set to 'All records match'. The 'Data Source' section at the bottom shows a list of data sources: Customer Satisfaction, Regional Sales Revenue, Sales Revenue by State & Area, Monthly Charge, Sales Revenue by State & Contr..., Contract Plan Subscription by R..., and Churn Rate Vs C.

#	churn_clean.csv	churn_clean.csv	churn_clean.csv	churn_clean.csv
Case Order	Customer id	Interaction	UID	
1	K409198	aa90260b-4141-4a24-8e36-...	e885b2	
2	S120509	fb76459f-c047-4a9d-8af9-e...	f2de8b	
3	K191035	344d114c-3736-4be5-98f7-c...	f1784cf	
4	D90850	abfa2b40-2d43-4994-b15a-...	dc8a36	
5	K662701	68a861fd-0d20-4e51-a587-8...	aabb64	
6	W303516	2b451d12-6c2b-4cea-a295-b...	97598f	
7	U335188	6630d501-838c-4be4-a59c-...	87d1c4	
8	V538685	70ddaa89-b726-49dc-9022-...	fce3f21	
9	M716771	05a49ee3-8fd5-453a-a5f3-8...	6c7043	
10	I676080	86f17e4d-2c24-4b70-a6ec-d...	6ffe183	
11	J980369	89490f4b-765f-431a-b302-5...	6271cc	
12	E243720	c32b8522-d62a-49b1-a77a-e...	073356	

- **four different data representations to summarize the data or display trends**

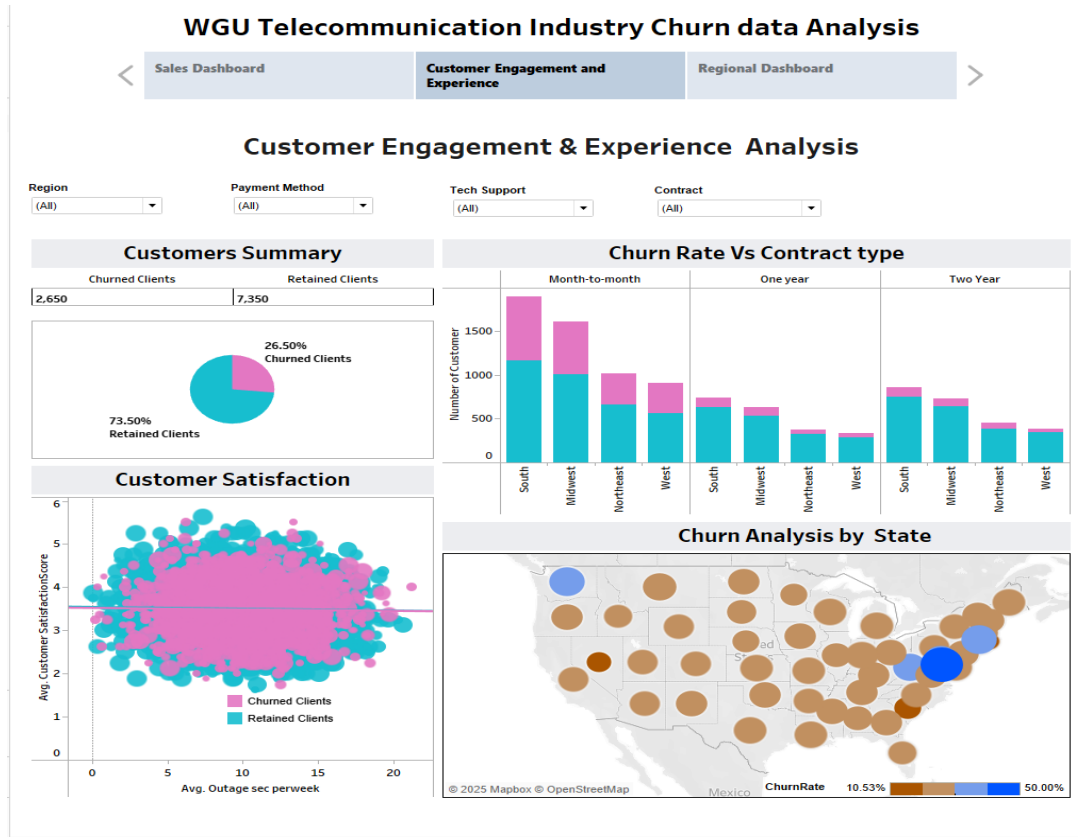
For the purposes of analysis, I have created three dashboards to cater to the expectations of the executive leadership:

- Customer Engagement and Experience Analysis Dashboard
- Sales Dashboard
- Region Analysis Dashboard

The visual representation per dashboard is outlined below:

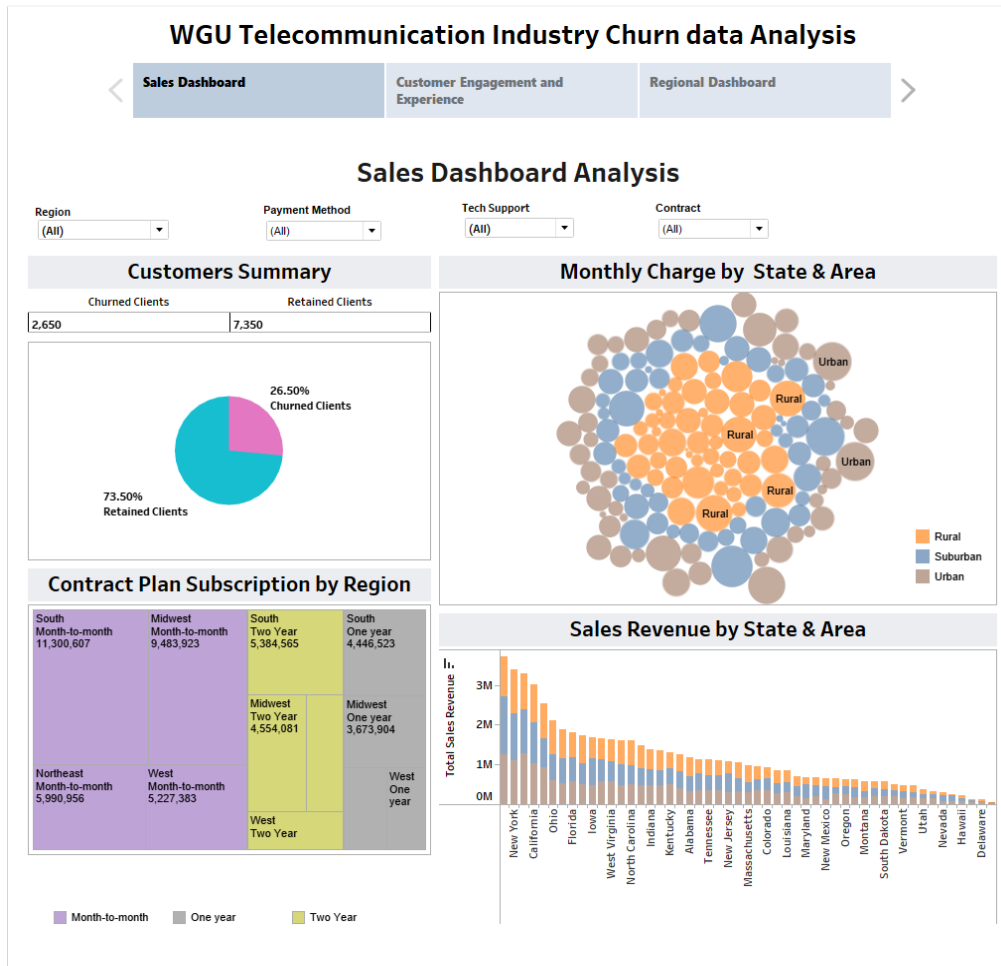
The Customer Engagement and Experience Analysis dashboard comprehensively overviews customer retention and satisfaction. The following data visualizations were created

1. Pie Chart: Summarizes the number of churned vs. retained customers for quick insights.
2. Scatterplot: Displays the relationship between average customer satisfaction scores and average weekly outage time with a trendline. Hovering over data points reveals additional details such as bandwidth, customer tenure, and monthly charges.
3. Map: Visualizes the churn rate by state, helping identify regional trends.
4. Stacked Bar Chart: Shows contract types by region, with color coding to represent churn, offering insight into contract-related churn patterns.



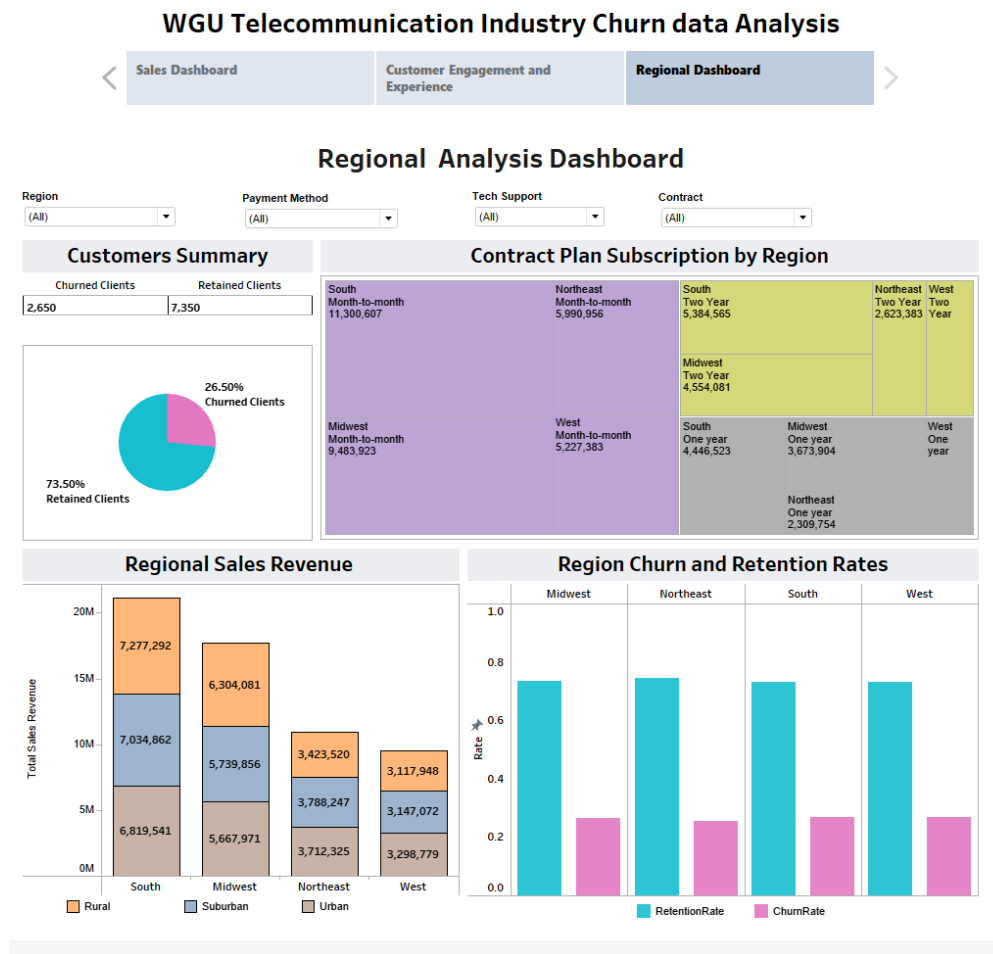
The Sales Dashboard provides key insights into customer behavior and sales performance. For the Sales analysis, I have created the following visualizations

1. Pie Chart: Summarizes churned vs. retained customers, with a tooltip displaying total sales revenue for a deeper understanding of customer impact.
2. Heatmap: Visualizes contract plan subscriptions by region, helping to identify geographic trends.
3. Bubble Chart: Displays monthly charges by state, with color-coding by area, offering insights into pricing variations.
4. Stacked Bar Chart: Shows sales revenue by state and area, highlighting the performance across different regions.



The Region Analysis Dashboard provides insights into customer retention and sales performance across different regions. For the regional VP, I have created the following dashboards,

1. Pie Chart: Summarizes churned vs. retained customers, with a tooltip displaying total sales revenue.
2. Heatmap: Shows contract plan subscriptions by region, with tooltips revealing sales figures.
3. Stacked Bar Chart: Visualizes sales revenue distribution across regions.
4. Side-by-Side Bar Chart: Compares churn and retention rates by region, with tooltips providing a breakdown by state.

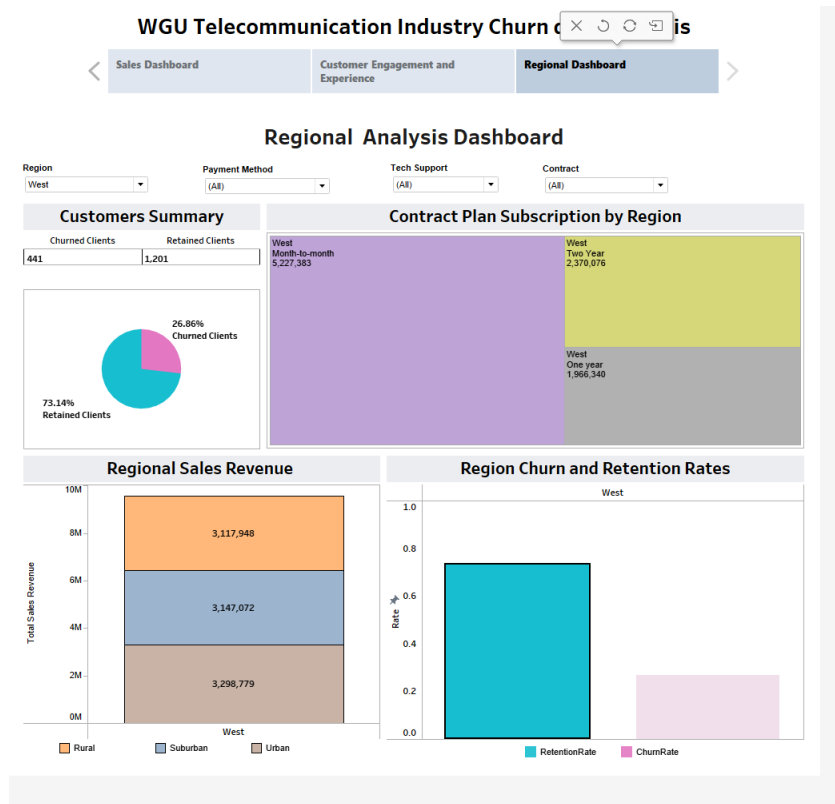
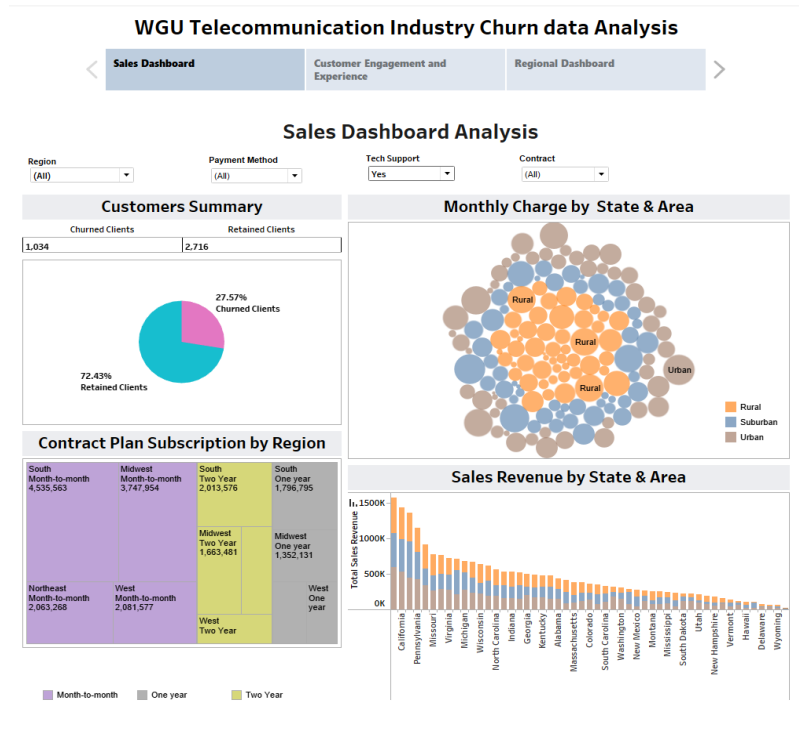


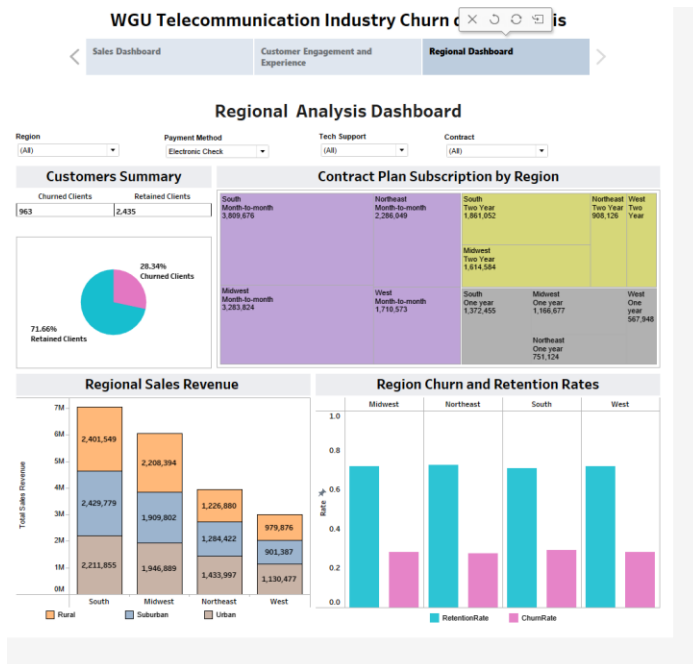
- two different interactive controls that allow the user to modify the presentation of data

For interactive controls, I have added slicers and interactive visuals that allow the user to modify the presentation of the data

1. Slicers and Filters

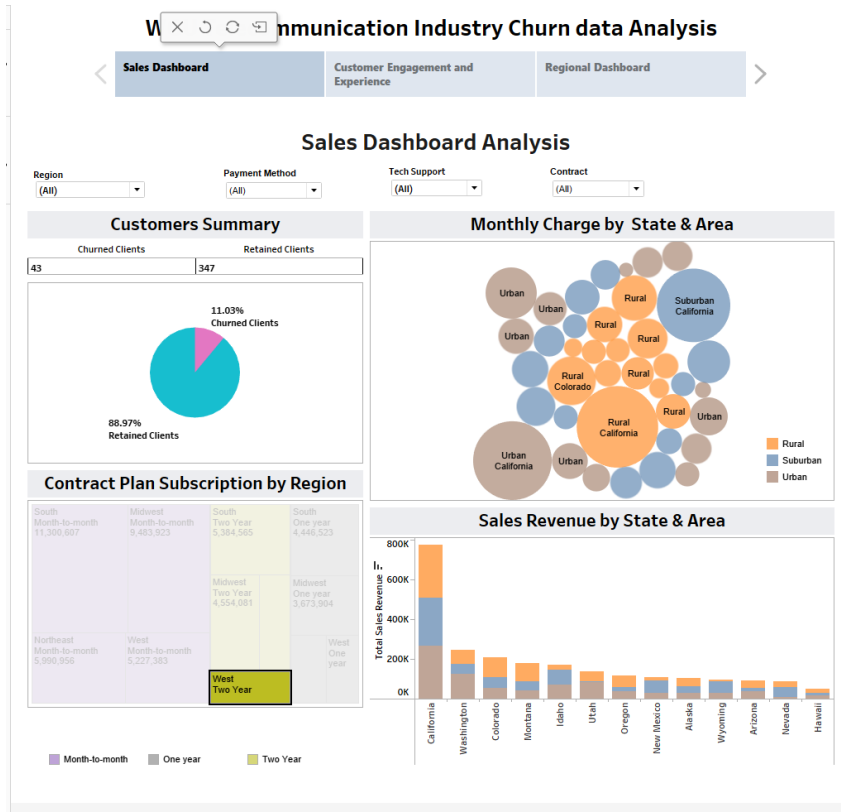
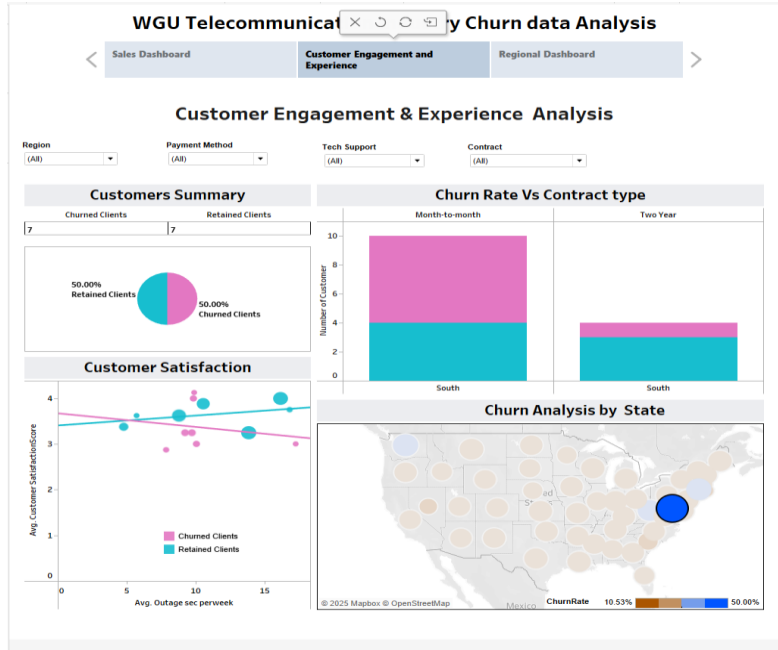
Interactive filters for region, payment method, tech support, and contract type allow for detailed analysis and trend identification.

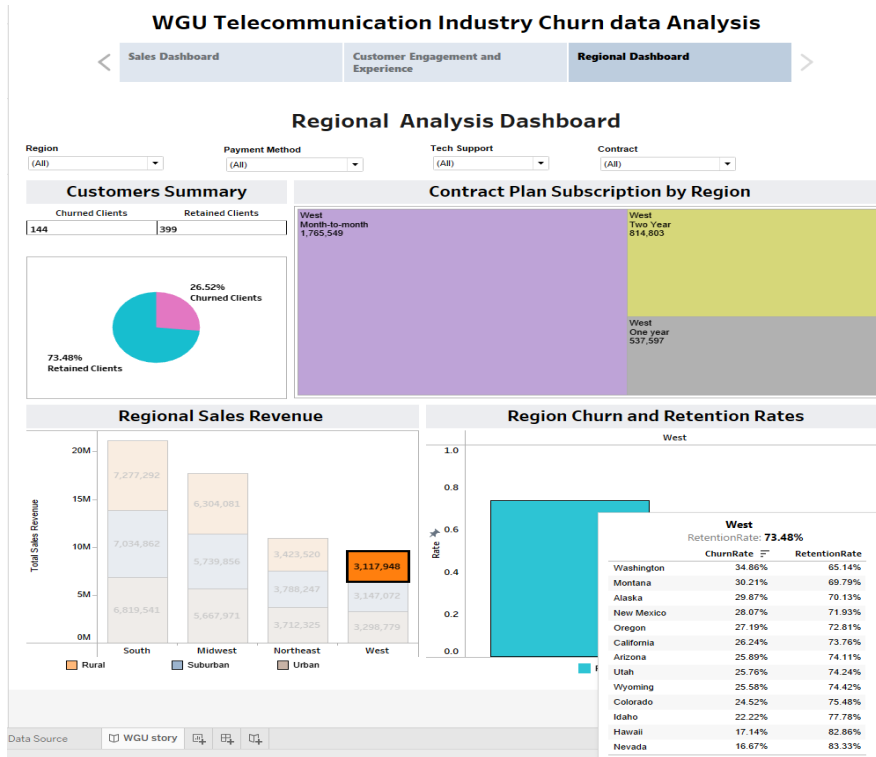




2. Interactive visuals

Interactive visuals enable users to explore data more deeply by clicking on elements within the visual. For example, in the Customer Engagement and Experience Analysis dashboard, if a user clicks on a specific state on the map, all other visuals (like the bar charts or pie charts) will update to show only data related to that state. This interactivity allows users to quickly adjust the scope of the data they analyze without applying traditional filters.





- two different metrics or key performance indicators computed using data from both chosen data sets

Using the data from the Customer Engagement and Experience Analysis Dashboard, Sales Dashboard, and Region Analysis Dashboard, two key performance indicators (KPIs) that were computed are:

- Churn Rate (%)

Formula:

$$\text{Churn Rate} = (\text{Number of Churned Customers} / \text{Total Number of Customers}) \times 100$$

This Churn rate KPI is crucial for understanding customer retention and identifying potential issues that may lead to customer dissatisfaction. By analyzing churn by region, contract type, and customer experience factors (e.g., outage seconds per week, satisfaction score), the

telecommunication industry can implement targeted strategies to improve retention. The Churn Rate by State (Map Chart) and Side-by-Side Bar Chart for Churn vs Retention Rate by Region provide insights into where churn is highest.

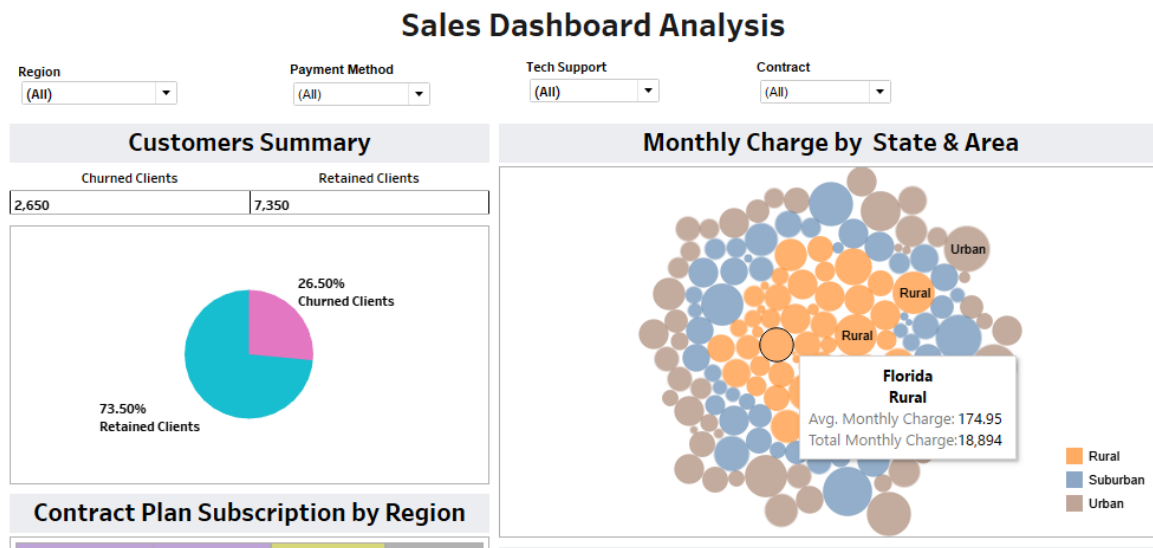
2. Average Monthly Charge

Formula:

Average Monthly Charge = SUM(Monthly Charges)/Total Number of Customers

This KPI helps assess the typical revenue generated per customer every month. The telecommunication industry can identify pricing trends and potential factors influencing customer retention by analyzing average monthly charges across different regions, contract types, and churn status. The Bubble Chart (Monthly Charge by State, Colored by Area) provides insights into how pricing varies geographically. The average monthly charge can be utilized alongside the churn rate to determine if higher charges correlate with customer dissatisfaction or retention.

Tracking churn rate alongside average monthly charge helps businesses evaluate pricing strategies and their impact on customer loyalty.



A2: DATA SETS

Data sets used

1. Telecomm Churn industry data set (churn_clean.csv)
2. Region and State data set (States.csv)



churn_clean.csv



states.csv

A3: INSTALLATION INSTRUCTIONS

Step-by-Step Dashboard Installation Guide

1. Prerequisites

Before installing the dashboard, ensure you have the following:

- A computer with Windows or macOS
- Tableau Desktop or Tableau Reader installed
- Access to the dashboard file (.twbx or .twb)



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- The CSV data files



states.csv



churn_clean.csv

2. Installing Tableau (If Not Already Installed)

- Visit the Tableau website (<https://www.tableau.com/>).
- Download Tableau Desktop (for full functionality) or Tableau Reader (for viewing only).
- Follow the on-screen instructions to install the software.
- Once installed, open Tableau.

3. Opening the Dashboard

- Locate the dashboard file (.twbx or .twb) on your computer.
- Double-click the file to open it in Tableau Desktop or Tableau Reader.

4. Connecting to Data (If required)

- Select the appropriate data source (CSV) if prompted.
- Verify that the data loads correctly into Tableau.

5. Verifying Installation

- Ensure all visuals and filters are displaying correctly.
- Test interactions such as clicking on charts and applying filters.
- If any issues arise, confirm that data sources are correctly linked and try refreshing the data.

A4:NAVIGATION INSTRUCTIONS

- **Overview**

The WGU telecommunication Industry Churn data analysis story is designed for intuitive and interactive analysis, helping executive leaders make data-driven decisions. Users can gain deep insights into customer behavior, sales trends, and regional performance by using the filtering options, exploring tooltips, and interacting with the visuals.

The interactive dashboard provides insights into customer engagement, sales performance, and regional trends. The story consists of three main sections:

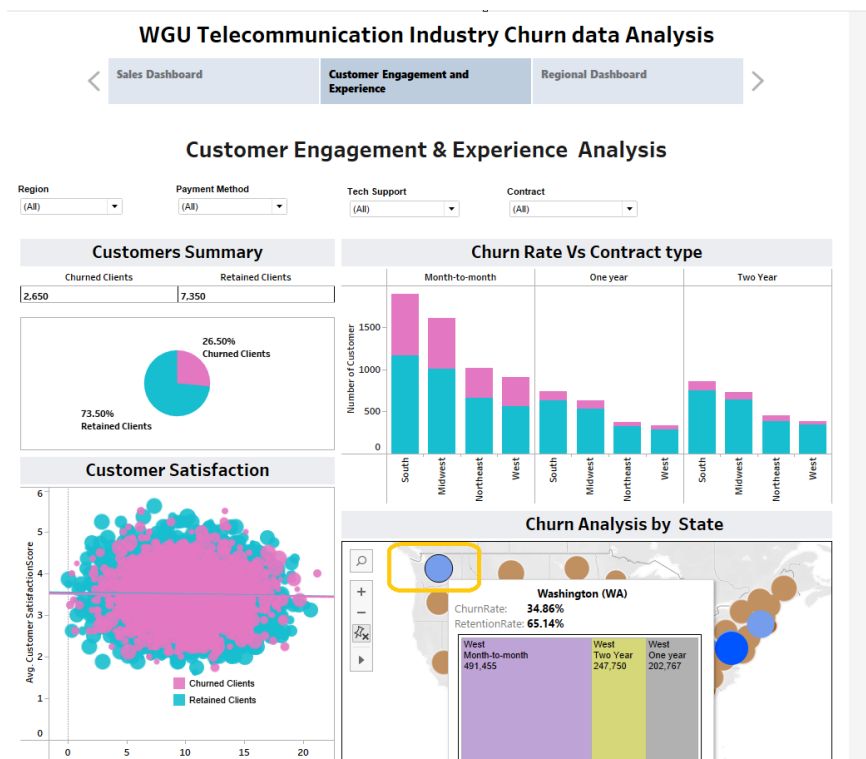
1. Customer Engagement and Experience Analysis Dashboard
2. Sales Dashboard
3. Region Analysis Dashboard

Each dashboard contains interactive visualizations that help users analyze key business metrics such as churn rate, customer satisfaction, contract types, and sales revenue.

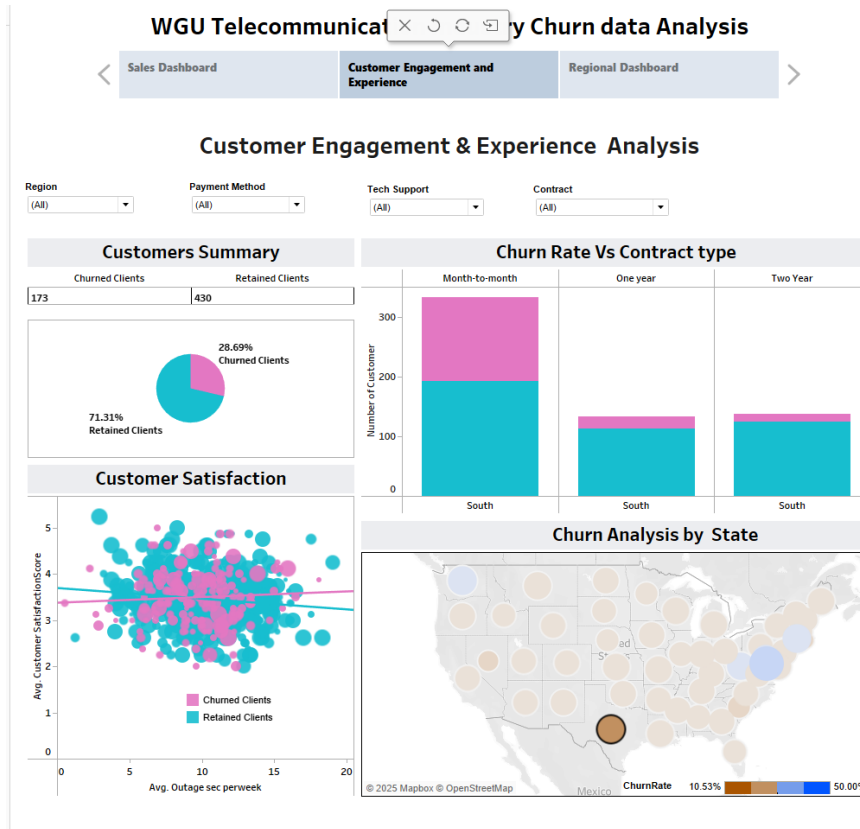
- **Navigating the Dashboards**

1. Interacting with Visuals

Hover Over Charts: Hover your cursor over any chart or data point to view additional details in the tooltip. Tooltips provide insights such as total sales revenue, churn rates, and other key metrics.



Click on Data Points: Clicking on elements within a chart (such as a specific region, contract type, or customer segment) filters other visuals in the dashboard to reflect relevant data.



2. Using Filters

Each dashboard includes the filters Region, Payment Method, Tech, and Contract. That allows users to refine data views.

To apply filters:

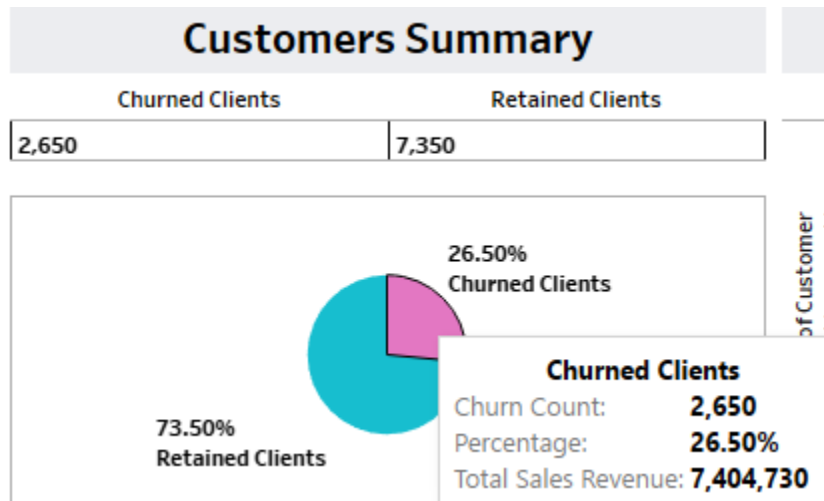
- Locate the filter panel on the side of the dashboard.
- Click on dropdown menus to select specific categories
- Select one or multiple filter options to update the visuals dynamically.
- Click apply



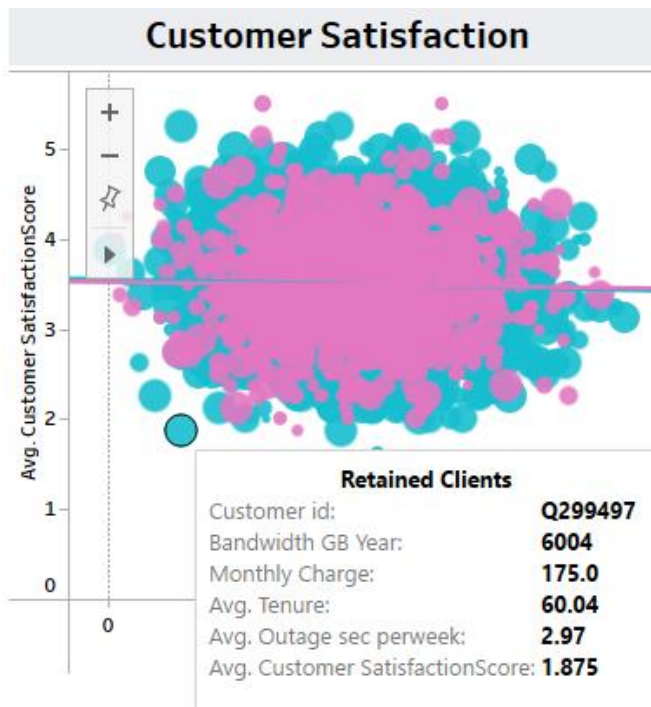
3. Understanding Key Visuals

- Customer Engagement and Experience Analysis Dashboard**

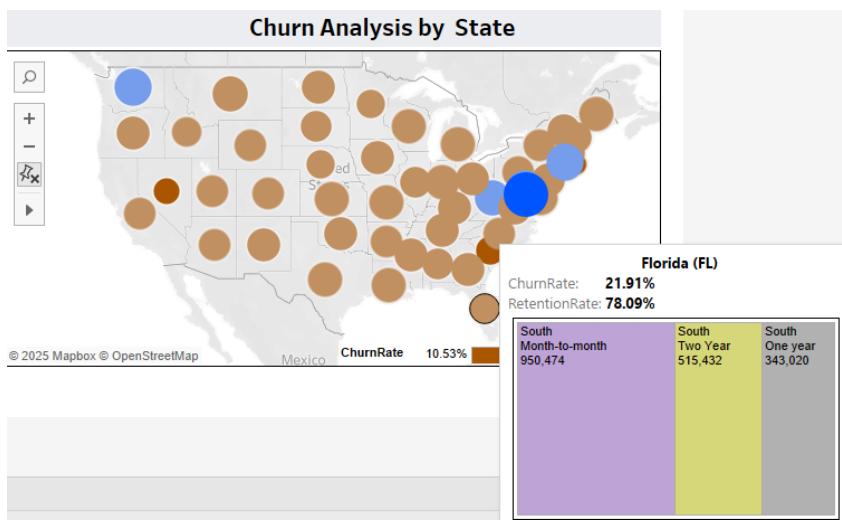
1. Pie Chart: Displays the percentage of churned vs. retained customers. With a tooltip that shows the count and the sales revenue



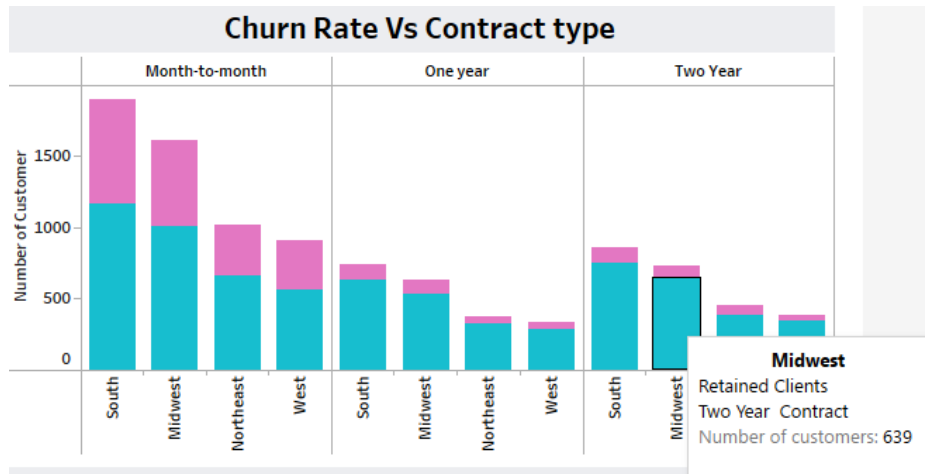
2. Scatterplot: The relationship between customer satisfaction and outage seconds per week with insights on bandwidth, tenure, and monthly charges added on the tooltip. The trendline shows if there is a correlation between customer satisfaction and the average outage seconds per week



3. Map: Illustrates churn rate by state.

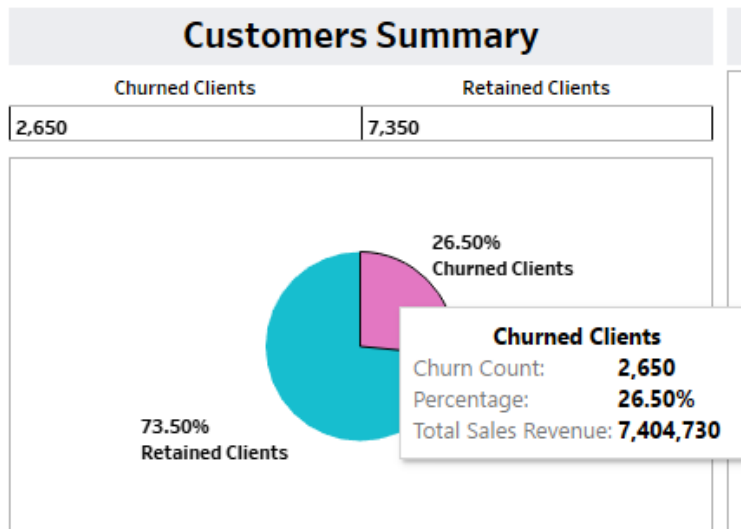


- Stacked Bar Chart: This breaks down contract types by region and highlights churn trends.

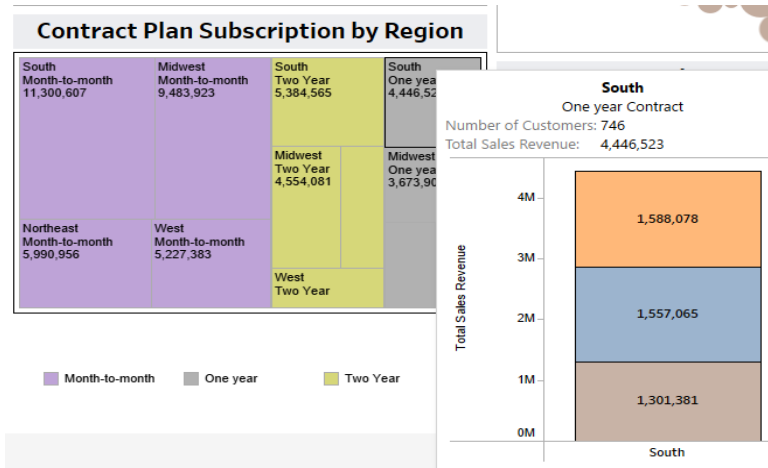


- Sales Dashboard**

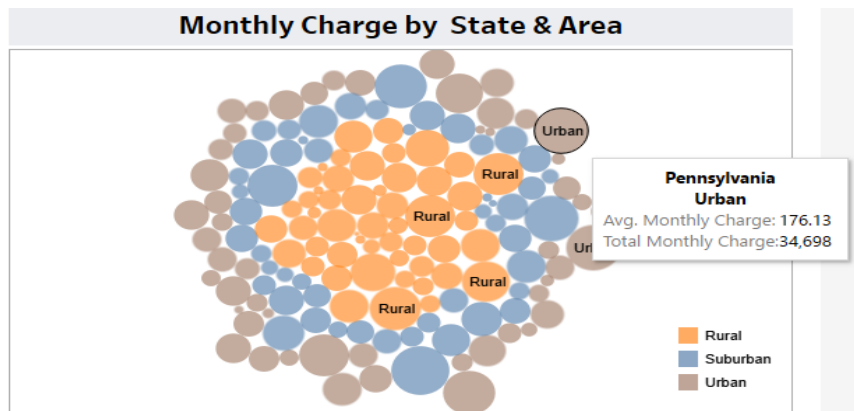
- Pie Chart: Summarizes churned vs. retained customers and includes a tooltip for total sales revenue.



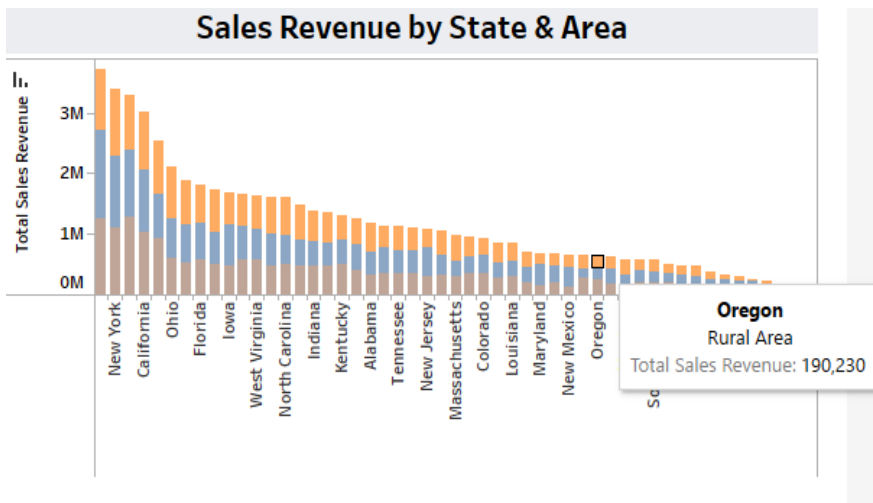
- Heatmap: Displays contract plan subscriptions by region with a tooltip that shows the total sales revenue for that region/contract type segmented by the area



- Bubble Chart: Represents monthly charges by state, colored by area. With a tooltip that shows the total monthly charge and the average monthly charge for that area

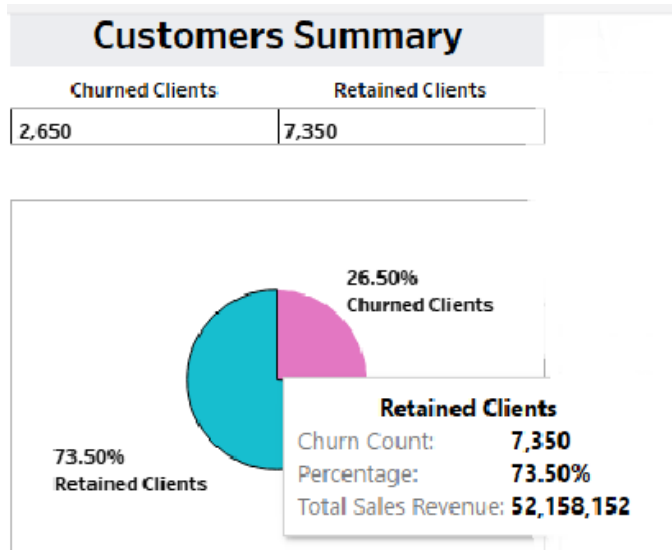


- Stacked Bar Chart: Showcases sales revenue distribution across states and areas.



- **Region Analysis Dashboard**

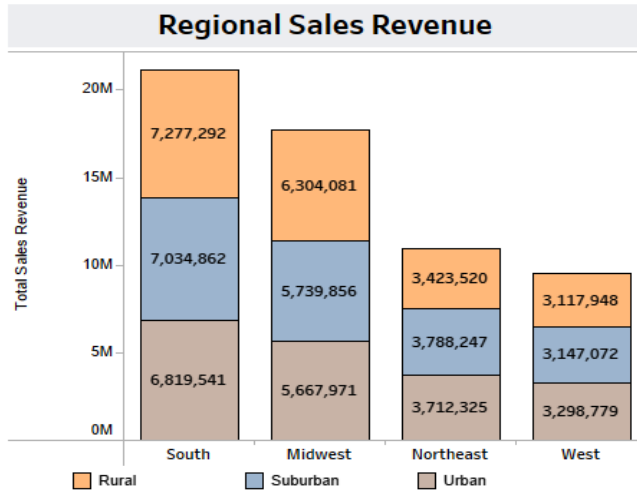
1. Pie Chart: Summarizes churned vs. retained customers with a tooltip for total sales revenue.



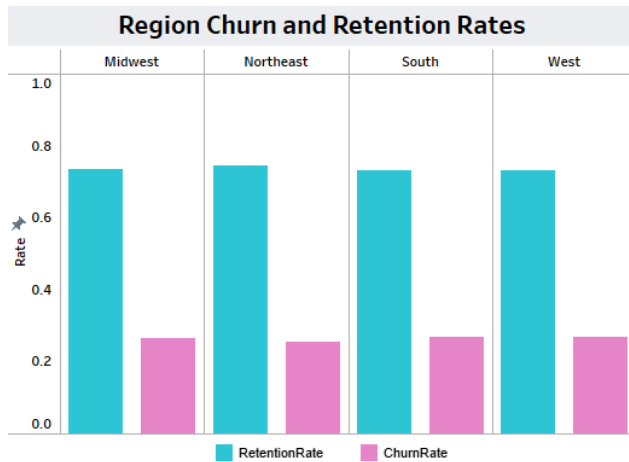
2. Heatmap: Displays contract plan subscription by region with sales insights.

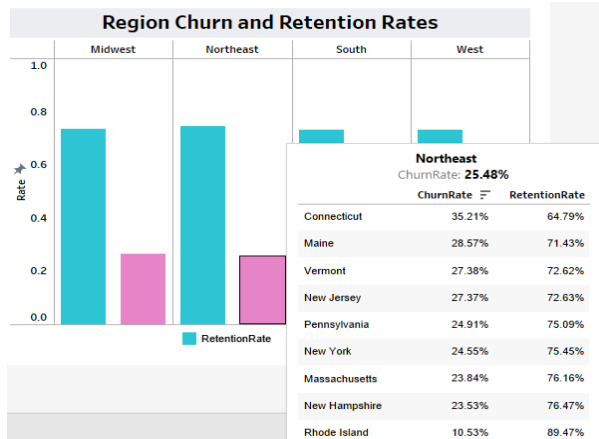
Contract Plan Subscription by Region					
South Month-to-month 11,300,607	Northeast Month-to-month 5,990,956	South Two Year 5,384,565		Northeast Two Year 2,623,383	West Two Year
		Midwest Two Year 4,554,081			
Midwest Month-to-month 9,483,923	West Month-to-month 5,227,383	South One year 4,446,523	Midwest One year 3,673,904	West One year	
			Northeast One year 2,309,754		

3. Stacked Bar Chart: Shows sales revenue by region.



4. Side-by-Side Bar Chart: Compares churn and retention rates by region, with a tooltip for churn and retention rates by state.



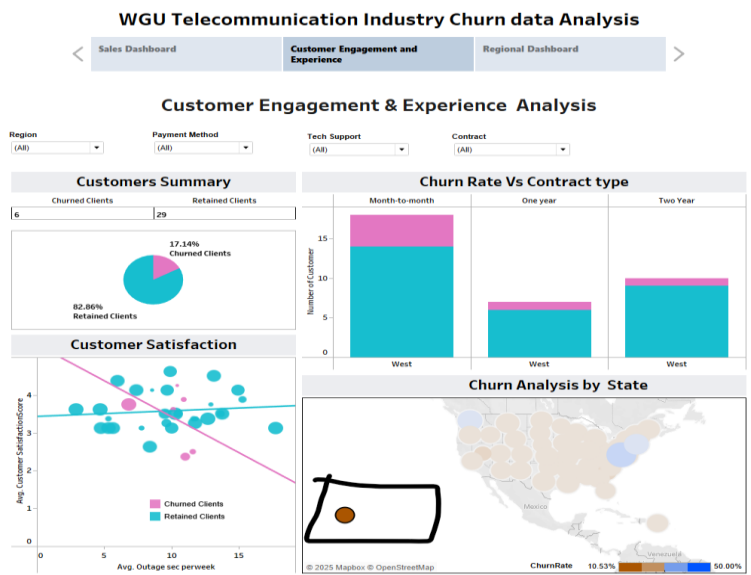


4. Resetting Filters

Click the Reset Filters funnel or manually deselect filters to return to the default view.



To reset an interactive visual, click on the same data point that was used to filter the data or click on a blank space on the visual



5. Navigating Between Dashboards

Use the tabs at the top of the dashboard to switch between the views (Sales dashboard, Customer Engagement and Experience dashboard, and Region Analysis dashboard).



B: PANOPTO STORYTELLING WITH DATA

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=efbc0a15-f4d0-421c-afcd-b2b1003a26fd>

C1: DASHBOARD ALIGNMENT

The WGU telecommunication industry Churn Dataset contains various attributes related to customer retention, contract details, sales performance, and regional distribution. I have combined the dataset with an external dataset for states and created dashboards that comprehensively overview key metrics, including churn rate, retention rate, subscription patterns by region, contract type and Area, and revenue distribution. The dashboards are designed to address the following needs from the data dictionary.

The senior vice president for customer service experience (SVP) is interested in a dashboard that makes up the customer and drives their customer behavior to address their needs, I have created a pie chart that visually represents the proportion of churned vs. retained customers, allowing stakeholders to assess churn severity quickly. The stakeholders can assess the churn severity by various factors such as the state, contract type, region, and if the customer had tech support, which has been enabled through interactive dashboards and filters. I have also created a scatter plot that evaluates if there is a linear relationship between the customer satisfaction rate and average outage seconds per week. The customer satisfaction rate is a calculated field that

includes the computation of the average rating for the survey. Metrics Timely response, Timely fixes, Timely replacements, Reliability, Options, Respectful response, Courteous exchange, and Evidence of active listening. The customer engagement and experience dashboard aligns closely with the needs outlined in the WGU Churn Dataset's data dictionary by offering clear visual representations of customer behavior.

The executive Vice president of Sales (EVP) is interested in the broad categorization of customers and how these demographics play out across regions since they are tasked with the strategic recruitment and the retention or renewal sales for the organization to be able to address the needs of the EVP, I have created a sales dashboard allows the EVP to be able to see the customer retention by regions, heat map highlights contract subscription counts and revenues across different regions, revealing which regions and areas engage more with specific contract types. Using the monthly Churn and the Tenure of the customers, I have calculated the revenue the company could derive from the customer based on the customer's history. The customer's monthly charges have also been computed for each state, and a bubble chart was created for each state by the Area. The monthly charge and the sales revenue are analyzed in terms of sales performance, which can be analyzed by factors such as region, state, contract, area, and state. The EVP can derive strategies to recruit and retain customers using this data.

The panel of regional vice presidents (Regional VP) needs to be able to set policies and manage operations in their region. To address the needs of the regional VP, I have created the regional churn and retentional rates side-by-side bar chart that effectively contrasts retention and churn rates per region. This visualization helps in evaluating customer loyalty and retention strategies for each region. The heat map highlights contract subscription counts and revenues across different regions, revealing which regions and areas engage more with specific contract types.

The stacked bar chart visually compares total revenue across regions segmented by urban and rural areas. This helps stakeholders identify high-performing regions and areas needing improvement.

The data analytic peers are interested in the data and other actionable recommendations backed by the evidence. To address their needs, the dashboards have ensured that effective data representation has been created to ensure that complex datasets are interpreted correctly and efficiently. The charts are chosen based on the type of data to improve comprehension, such as pie charts for proportions, heat maps for density, bar charts for comparison, bubble charts for distribution, maps for geographical insights, and scatter plots for correlation analysis. The data visualizations directly reflect the dataset attributes outlined in the data dictionary, ensuring precise reporting. The insights derived from the dashboard will help decision-makers implement targeted strategies to reduce churn, increase retention, and understand the product penetration rates in various regions. Stakeholders can quickly extract insights from visual reports instead of analyzing raw tables, improving decision-making speed. They can also dice and slice the data on different metrics via the filters to obtain different perspectives of the same dataset.

The dashboard aligns closely with the needs outlined in the WGU Churn Dataset's data dictionary by offering clear visual representations of customer behavior, subscription patterns, and revenue distribution. Through thoughtful data representation and reporting, WGU Telecommunication can identify key trends, optimize strategies, and drive better business outcomes. By leveraging dashboards, the business can gain a strategic advantage in managing customer retention and maximizing revenue growth.

C2: ADDITIONAL DATA SET INSIGHTS

The additional state data enhances insights from the original churn dataset by providing a regional geographical and demographic perspective on customer behavior, service adoption, and sales trends. By incorporating the state's data, the analysis moves beyond overall trends and into region-specific strategies, enabling tailored customer engagement, optimized sales approaches, and improved service delivery based on geographic patterns. Here's how these variables contribute to a deeper analysis:

1. Regional Churn and Retention Patterns

The Churn Rate by State (Map) and Side-by-Side Bar Chart (Churn and Retention Rate by Region) leverage the Region variable to show where customer losses are highest and lowest. This allows executives to compare regions and identify factors contributing to churn, such as service quality or local competition.

2. Contract and Service Adoption by Region

The Heatmap (Contract Plan Subscription by Region) and Stacked Bar Chart (Contract Type by Region, Colored by Churn) help visualize how different areas prefer specific contract plans. The Tech Support, and Area variables offer additional insights into service adoption trends across different regions.

3. Revenue and Sales Performance Across Regions

Total Sales Revenue (Pie Chart Tooltip), Monthly Charge by State (Bubble Chart), and Sales Revenue by Region (Stacked Bar Chart) use the Region and Area variables to highlight differences in revenue generation.

This enables the EVP of Sales to tailor marketing and pricing strategies to maximize profitability in high-revenue regions and improve performance in underperforming areas.

4. Customer Satisfaction and Outage Impact by Region

Average Outage Seconds per Week and Customer Satisfaction Score are analyzed with the Region variable to assess whether certain areas experience more service disruptions.

This insight helps the SVP of Customer Experience focus on infrastructure improvements in highly dissatisfied regions.

5. Targeted Decision-Making for Regional Leadership

The Regional VPs can use these insights to adjust policies, marketing strategies, and service improvements based on their region's specific needs and behaviors.

The interactive dashboards ensure decision-makers can filter and drill down into region-specific data, providing actionable insights for that region.

C3: DECISION-MAKING SUPPORT

Effective organizational decision-making requires a comprehensive understanding of customer behavior, revenue trends, and regional performance. The dashboards provide various visualizations to support executive leaders in making informed decisions. Two key data representations from the dashboards that can significantly aid decision-making are the

1. Scatterplot: Customer Satisfaction vs. Average Outage Seconds per Week

The scatterplot visualizes the relationship between the average customer satisfaction score and average outage seconds per week, with a trendline highlighting the overall pattern. Additionally, tooltips provide further insights into bandwidth usage, tenure, and monthly charges for each data point. The visual can be sliced and diced using various metrics, such as Region, payment method, Tech support, and Contract.

This visualization enables executive leaders to:

- **Identify Service Quality Impact:** A clear negative trend between outage seconds and satisfaction may indicate a need for improved infrastructure investments.

- Segment High-Risk Customers: Targeted retention strategies can be developed by analyzing outliers (customers with high outage times and low satisfaction).
- Optimize Service Offerings: If customers with higher bandwidth usage and tenure have lower satisfaction, tailored solutions (e.g., premium support plans) can be introduced to enhance retention.

2. Stacked Bar Chart: Sales Revenue by State/Area

The stacked bar chart breaks down sales revenue by state/area, allowing executives to see revenue distribution across different geographic regions. The visualization can be sliced by various Metrics such as Region, payment method, Tech support, and Contract to aid in decision-making.

This visualization enables executive leaders to make decisions to

- Recognize High-Performing Regions: States or areas with the highest revenue can be further analyzed to understand best practices that can be replicated elsewhere.
- Address Underperforming Areas: Identifying states or areas with lower revenue allows executives to investigate potential causes, such as competitive market presence or lack of effective marketing strategies.
- Allocate Resources Efficiently: Based on revenue performance, resources such as marketing budgets, sales personnel, and promotional campaigns can be better allocated to maximize growth.

By leveraging these two visualizations, executive leaders can take a data-driven approach to decision-making. The scatterplot helps improve customer experience and retention strategies,

while the stacked bar chart enables revenue optimization across different regions. Together, these insights support strategic actions that enhance overall business performance

C4: INTERACTIVE CONTROLS

I added filter slicers and created interactive visuals to allow users to have interactive controls with the dashboards. These interactive controls allow users to filter and explore data dynamically, enabling a customized and engaging dashboard experience.

1. Slicers

Filter slicers in the dashboard are Region, Payment Method, Tech Support, and Contract Type. Slicers are visual elements on the dashboard that allow users to filter data interactively. Slicers allow users to filter data across the entire dashboard based on their selected criteria, simultaneously controlling what data is displayed across multiple visuals. Users can dynamically adjust the data across all connected visuals by selecting specific criteria. When a user selects a value from a slicer, such as a particular region, all the visuals on the dashboard are updated to reflect only the data related to that region. For instance, in the Sales Dashboard, selecting a specific region in the slicer will filter the pie chart, heatmap, and other visuals to display data relevant only to that region. This allows users to drill down into specific subsets of data and view more focused insights.

2. Interactive Visuals (Visuals that Filter Others)

Interactive visuals provide a more focused, on-the-fly way for users to filter data by interacting directly with specific data points in the visuals. Clicking on a visual element updates the dashboard to show related data, offering a seamless and intuitive user experience. When a user clicks on a data point (for example, a segment of a pie chart or a region on a map), it triggers a

filter that updates the other visuals on the dashboard to display data corresponding to that selection. Interactive visuals enable users to explore data more deeply by clicking on elements within the visual. For example, in the Customer Engagement and Experience Analysis dashboard, if a user clicks on a specific state on the map, all other visuals will update to show only related data. This interactivity allows users to quickly adjust the scope of the data they analyze without applying traditional filters.

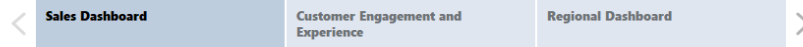
C5: COLOR BLINDNESS

To ensure the dashboards are accessible to colorblind individuals, I implemented several best practices in color selection, contrast, and alternative data representation. By using a colorblind-friendly palette, high contrast, alternative data representation methods, and interactive tooltips, the dashboard is accessible to all users, regardless of their color perception abilities. These accessibility enhancements ensure that the insights provided by the dashboard remain clear and useful to everyone. The following strategies were used to make the visuals clear and usable for all users, regardless of their ability to perceive certain colors.

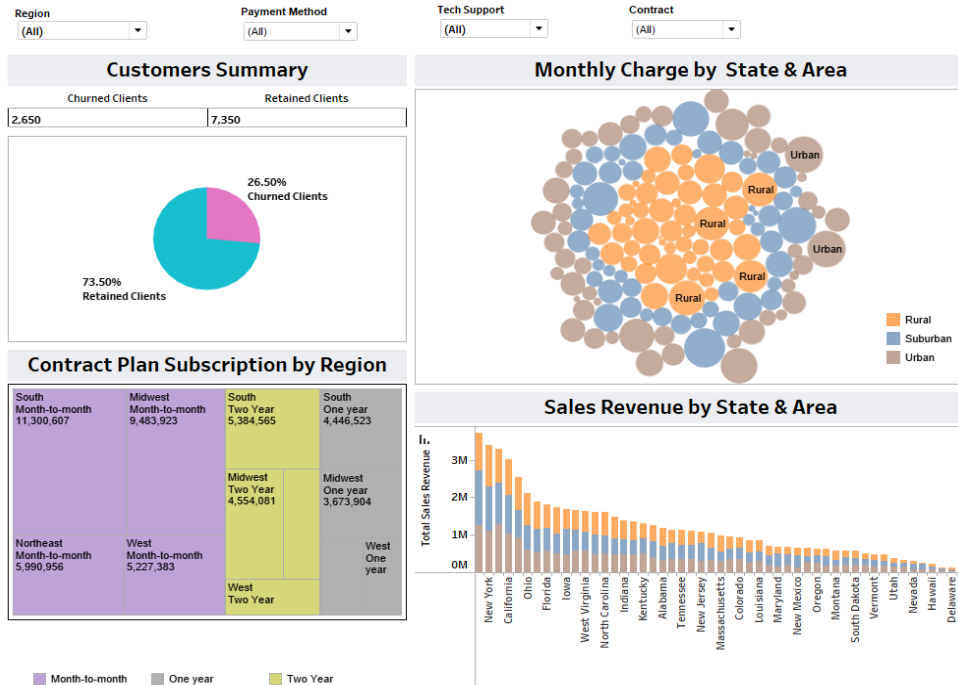
1. Colorblind-Friendly Palette

- I selected a color palette that accommodates various types of color blindness (protanopia, deuteranopia, and tritanopia). The colors used in the visuals for area categories. For the area variable, I used blue, orange, brown for the churn rate, used blue and pink and purple, gray, and yellow-green to differentiate the different types of contract

WGU Telecommunication Industry Churn data Analysis



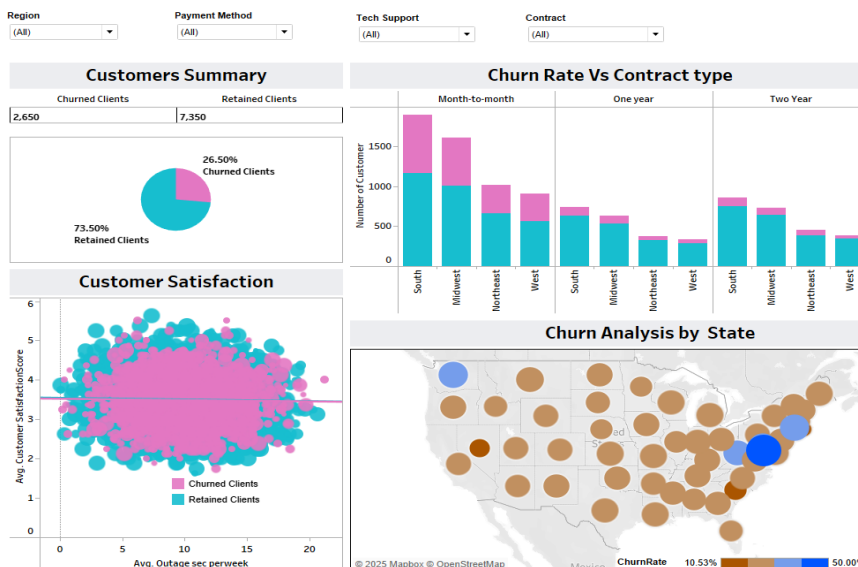
Sales Dashboard Analysis

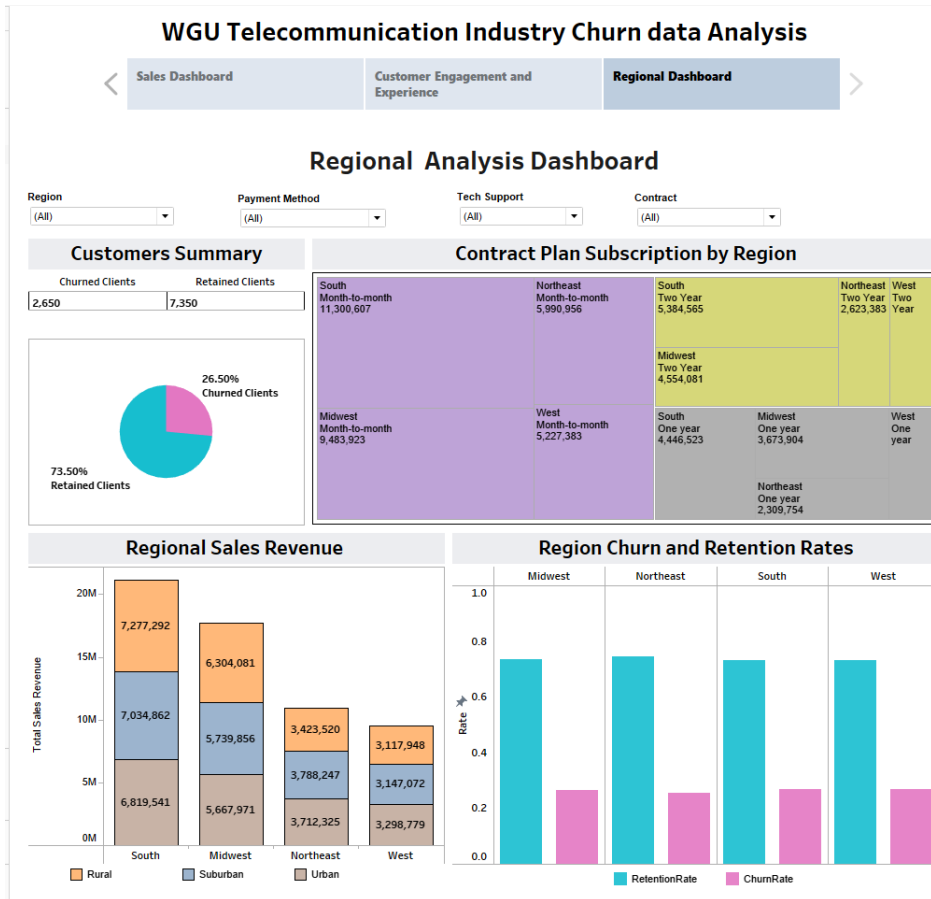


WGU Telecommunication Industry Churn data Analysis



Customer Engagement & Experience Analysis





- High-contrast combinations like blue, orange, and brown were chosen to differentiate categories.
- I avoided red-green and other problematic combinations that are difficult for colorblind users to distinguish.
- I used shades of blue and orange, as these are easier to differentiate for most colorblind individuals.

2. Alternative Data Representation

In addition to using color, I incorporated alternative ways to distinguish data:

- Labels and annotations were used extensively to clarify key information, reducing reliance on color alone.

3. High Contrast and Legibility

- Ensured strong contrast between foreground and background elements to improve readability.
- Tooltips provide hover-based details, allowing users to see specific information without relying on color differentiation.
- The text and numerical labels were adjusted for clear visibility and did not blend into the background.

4. Interactive Features for Clarity

Since all visuals are interactive and part of a user story, users can:

- Hover over elements to see additional data through tooltips, ensuring they get the information they need regardless of color perception.
- Use filters to break down the data into smaller, easier-to-analyze segments without relying on color differences alone.

C6: DATA REPRESENTATIONS

The two data representations that support the story and provide a plausible reason to use the Churn dataset and the state's dataset are

- The heatmap (Contract Plan Subscription by Region)
- side-by-side bar chart (Churn and Retention by Region)

The heatmap (Contract Plan Subscription by Region) and the side-by-side bar chart (Region Churn and Retention rates) are critical in conveying the data-driven story within the presentation.

The heatmap offers a geographical perspective on contract plan preferences. At the same time, the bar chart provides a comparative analysis of churn and retention rates, enabling executive

leaders to make informed strategic decisions. These visuals collectively support enhancing customer experience, improving retention, and optimizing sales strategies.

1. Heatmap: Contract Plan Subscription by Region

The heatmap visually represents contract plan subscription trends across different regions, using color intensity to indicate areas with higher or lower subscription rates. The heatmap utilizes a dataset from the Churn dataset(Contract variable) and the state's dataset (region variable)to tell the story. This visualization is essential in helping the users identify patterns in customer preferences and developing targeted sales strategies and engagement initiatives. The tooltip feature, which displays sales data, enhances the depth of insights by showing how contract subscriptions correlate with revenue performance.

The heatmap can be used to understand customer preferences and refine retention strategies for regions with lower contract subscription rates; it can also be used to analyze which contract plans perform best in different regions and adjust marketing and sales efforts accordingly; it can also be used to implement region-specific promotions, improving customer acquisition and retention. It can be used to explore the relationship between contract plans and churn rates, identifying areas for improvement. The heatmap highlights regional customer behaviors, allowing leaders to make informed decisions about product offerings and engagement strategies.

2. Side-by-Side Bar Chart: Churn and Retention Rate by Region

The visualization presents a comparative analysis of churn and retention rates across different regions, offering a clear picture of customer retention challenges. The Side-by-Side Bar Chart utilizes the Churn dataset(Churn variable) and the State's dataset (region variable)to tell the story. By displaying churn and retention rates, the chart lets stakeholders quickly identify which

regions struggle with customer retention. Additionally, the tooltip shows churn and retention rates by states within each region and provides granular insights for more targeted interventions. This data representation is crucial because it can be used to develop customer retention initiatives in high-churn regions, show how sales efforts correlate with retention rates, and refine customer outreach strategies. It can be used to implement localized policies that address churn challenges, allow for investigation of factors contributing to churn, and suggest data-driven solutions to reduce customer attrition. By providing a regional breakdown of churn and retention, this visualization helps the organization take proactive measures to improve customer loyalty and engagement.

C7: AUDIENCE ANALYSIS

In preparation for the story for different audiences, I considered their roles, areas of expertise, and the key decisions they needed to make based on the data. Here's how I used audience analysis to adapt the dashboard:

1. Senior Vice President for Customer Experience (SVP)

The customer experience dashboard focuses on increasing customer engagement, retention, and understanding customer behavior. Based on this, the dashboard was prepared to

- Provide Customer-Centric Insights whereby I highlighted key customer-related insights, focusing on churned and retained customers to help the SVP understand factors influencing customer behavior. The pie chart summarizing churned vs. retained customers allows for an immediate high-level overview of engagement.
- Provide satisfaction and outage correlation. I utilized the scatter plot to show the relationship between the average customer satisfaction score and average outage time per week with a trendline to help the SVP identify areas affecting satisfaction and retention.

- Provide geographical insights. I created a map to show the churn rate by state, and the stacked bar chart of contract types by region helps the SVP spot regional patterns that could influence customer engagement strategies.

The visuals have tooltip Information, and by including tooltips with information such as bandwidth, tenure, and monthly charge, the SVP can dive deeper into customer characteristics that might drive churn or retention.

2. Executive Vice President of Sales (EVP)

The sales dashboard was created to provide actionable insights on strategic recruitment, sales, and customer retention. To be able to accomplish this, the dashboard incorporated,

- Sales-Oriented Insights: The pie chart summarizing churned and retained customers is paired with a tooltip showing total sales revenue, which is important for the EVP to assess how customer retention and churn directly impact sales performance.
- Geographical Trends: The bubble chart that shows monthly charges by state, colored by area, and the stacked bar chart that shows sales revenue by state/area provide regional insights that help the EVP understand how different regions are performing in terms of both customer retention and sales revenue.
- Demographic & Contract Insights: The heatmap showing contract plan subscriptions by region gives the EVP insights into which contract types are most popular across different regions, assisting in decision-making about product offerings or sales strategies.

3. Regional Vice Presidents (Regional VP)

The focus of the regional vice president is to manage operations and set policies based on regional needs and performance. To be able to accomplish this, the dashboard incorporated,

- **Regional Performance Metrics:** The side-by-side bar chart showing churn and retention rates by region with a tooltip that provides churn and retention rates by state allows Regional VPs to understand regional dynamics in customer retention.
- **Sales and Customer Engagement:** The stacked bar chart showing sales revenue by region and the heatmap that links contract plan subscriptions with sales provide Regional VPs with actionable insights to fine-tune local customer engagement and retention strategies.

4. Data Analytics Peers

In preparing the dashboards, the focus for data analytics peers is understanding the design, methodology, and results of data analysis, with a keen interest in actionable insights and recommendations. To be able to accomplish this, the dashboard incorporated,

- **Detailed Data Visualizations:** Since my peers are technical, I included interactive and detailed charts like the scatter plot with a trendline and tooltips, allowing for deep dives into the data and the relationships between different variables such as customer satisfaction, outage times, and churn.
- **Tooltips and Methodology Transparency:** I ensured that tooltips provided more granular data (like bandwidth, tenure, and monthly charge) to enable my peers to critically analyze the data and its methodology, allowing them to identify trends or areas for further exploration.
- **Regional Insights:** For my technical peers, I presented regional analyses in the form of maps, bar charts, and heatmaps to show the granular impact of policies and customer behavior at the regional level, helping them identify potential areas of improvement from a data perspective.

The dashboards were designed with each audience's needs in mind. I focused on clear, high-level insights and visual clarity for non-technical leaders. I ensured interactivity and depth for my technical peers, allowing them to explore the data further. Each dashboard component, from the customer retention pie chart to the regional sales revenue stacked bar chart, was designed to align with each group's unique focus and decision-making needs. This approach ensures that every stakeholder gets the most relevant and actionable insights from the data, enhancing their ability to make informed decisions.

C8: UNIVERSAL ACCESS

I have created a clear, accessible presentation that can be understood by all audiences, regardless of their technical knowledge or background. The following steps outline how I designed the presentation for universal access:

1. Layout and Visual Clarity

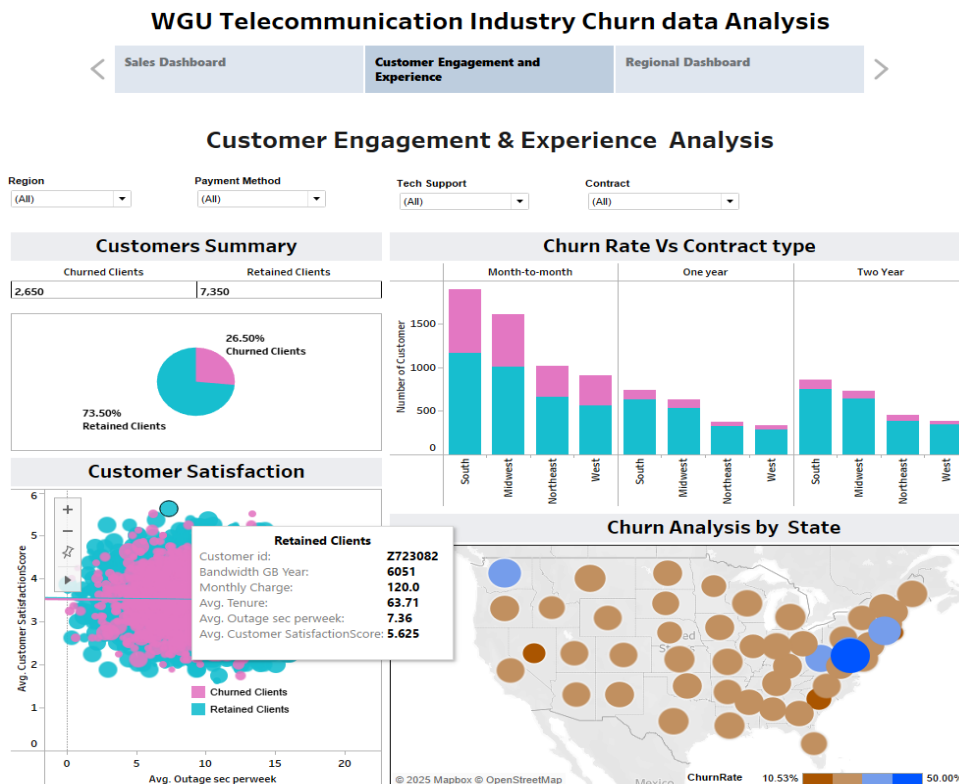
I organized the dashboards into distinct sections with easily identifiable charts that present key information in a structured manner. Each dashboard contains various visual elements with complementary designs, ensuring the audience can navigate easily between visuals. Key features include:

- **Pie Chart:** Summarizes the churned and retained customers. This is a familiar and intuitive visual, and the tooltip provides extra context (e.g., total sales revenue) when hovered over, ensuring clarity without overcomplicating the chart.
- **Scatterplot:** Displays the relationship between the average customer satisfaction score and the average outage time per week. This visual also includes a trendline and tooltip, offering relevant information such as bandwidth, tenure, and monthly charges. The tooltip's clarity ensures that users can grasp detailed data without overwhelming them.

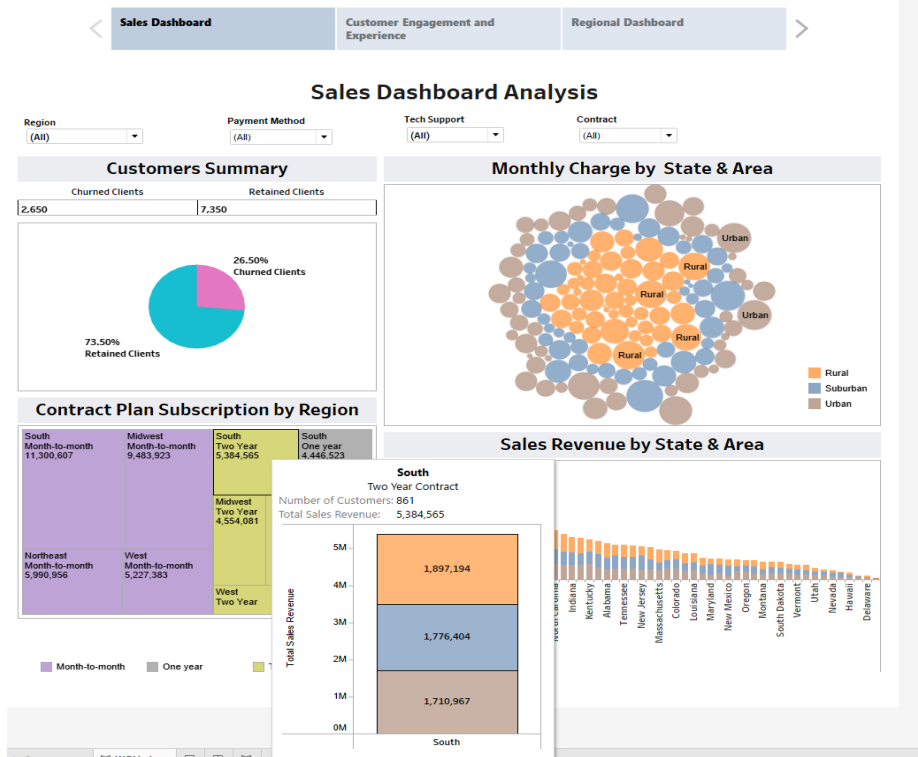
- Map: The churn rate is visualized by state, and color coding ensures that differences in churn rate are immediately apparent. It's designed for easy interpretation by anyone familiar with geographic maps.
- Stacked Bar Charts: These are used throughout the dashboards to present variables like contract types, sales revenue, and churn/retention rates. The color coding for churn status helps distinguish between the data categories clearly.

2. Accessibility Features

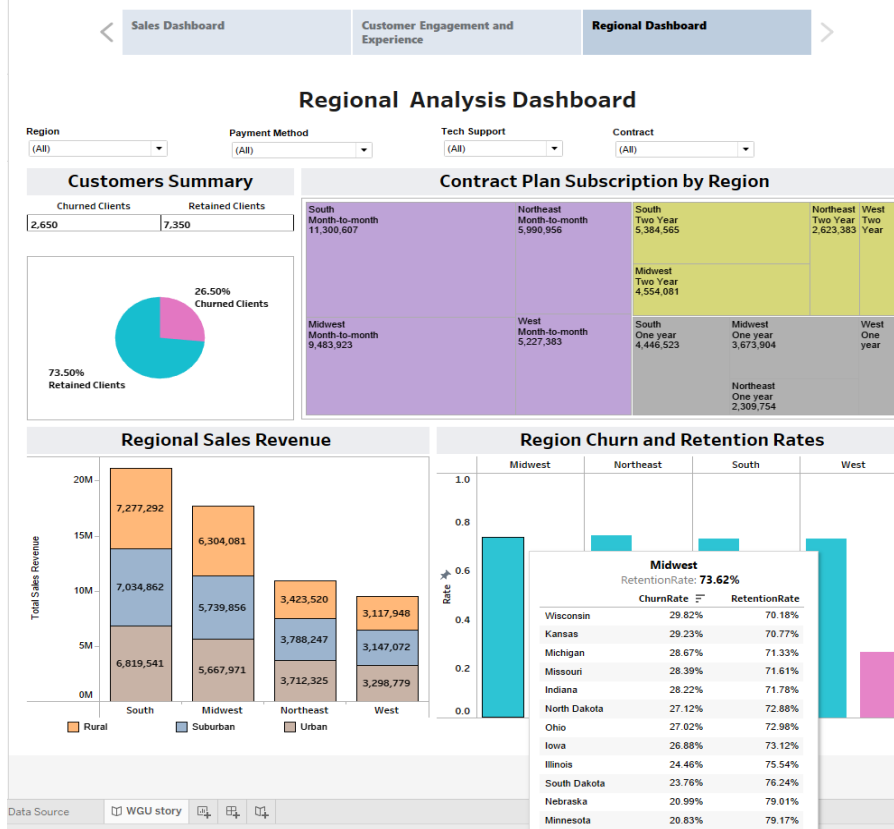
- Tooltips: All charts include tooltips that provide additional information when hovered over. This makes the dashboards accessible to beginners and more advanced users, enabling them to explore the data in detail without feeling overwhelmed.



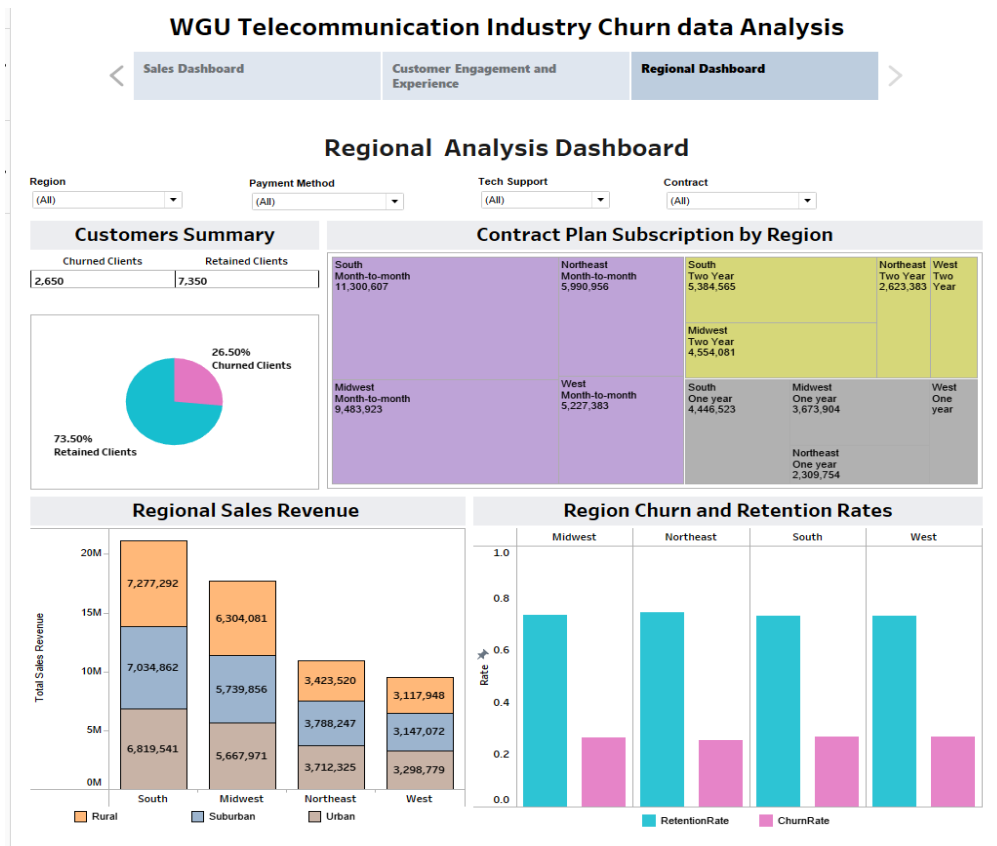
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- **Color Contrast:** I ensured that all visuals have sufficient contrast between colors, making it easier for users with visual impairments (e.g., color blindness) to distinguish between different categories (e.g., churned vs retained customers). The tooltips and color-coded data points further enhance accessibility.

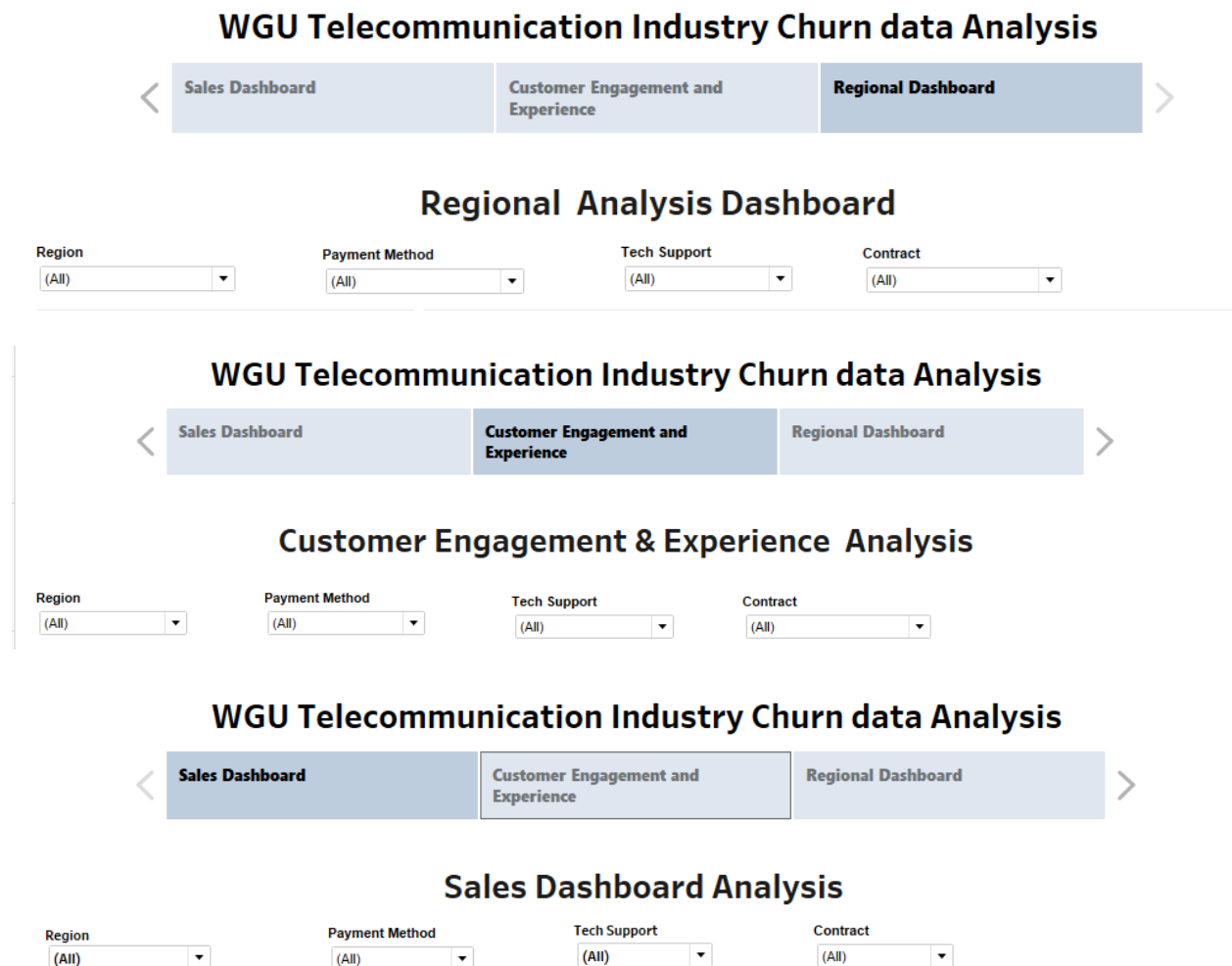


- **Text Descriptions:** Each dashboard has been paired with clear, concise descriptions for each visual. This text explains the data shown, the significance of the visuals, and how to interact with the dashboard. This approach helps people who may not be as familiar with the data presented.

3. User-Friendly Navigation and Filtering

Each dashboard allows users to interact with the visuals using Region, Payment Method, Tech Support, and contract filters. I designed the filters to be simple and intuitive, enabling users to

customize the data they view based on their preferences and needs. The filters and interactive features work seamlessly with the visuals, providing a cohesive experience that doesn't require any technical expertise.

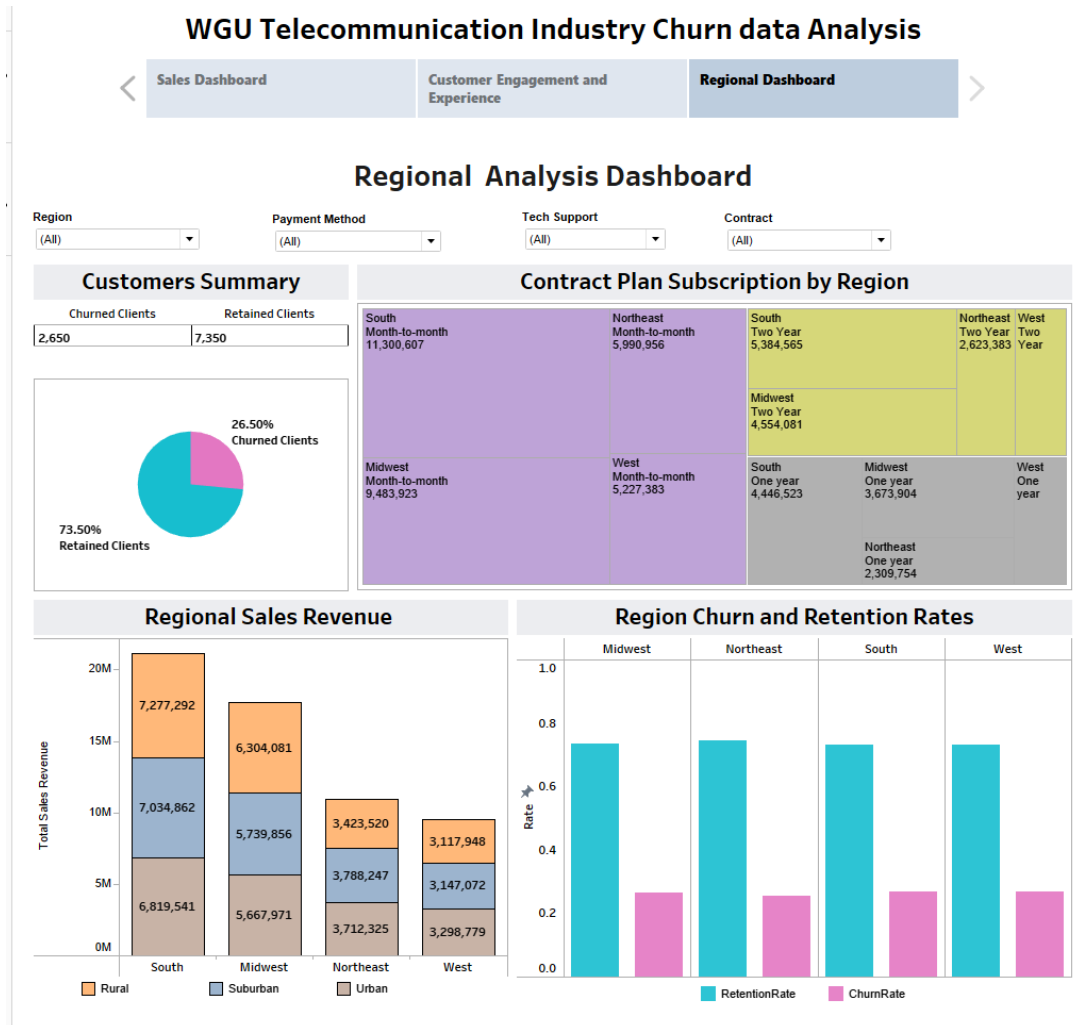


4. Consistent Design Across Dashboards

To make the presentation easy to follow, I used consistent design elements across all dashboards:

- The colors represented the same data categories (e.g., churned vs. retained customers, sales revenue) throughout.
- Font choices and sizes were selected for clarity, ensuring the text is legible, especially for people with visual impairments.

- The placement of filters is uniform across all dashboards, allowing users to navigate between sections easily.



5. Accessibility for All Users

Finally, I made sure that the design and presentation can be used by individuals with disabilities, following the principles of universal design. Features like tooltips, consistent colors, and clear navigation are intended to create an inclusive experience for all users, regardless of their experience level or physical abilities.

The design of this presentation ensures that all audiences, including those with visual impairments or minimal technical expertise, can engage with the data in a meaningful and

accessible way. The clear layout, interactive features, and tooltips provide all the users an intuitive and inclusive experience.

C9: EFFECTIVE STORYTELLING

1. Heatmap: Contract Plan Subscription by Region

The heatmap provides a visual representation of contract plan subscriptions across regions, color-coded to highlight areas with the highest concentrations of customers. This data visualization supports the story by showcasing each region's most popular contract plans, helping the SVP for Customer Experience understand customer preferences and the EVP of Sales refine marketing strategies. Additionally, the heatmap's tooltip, which includes sales data, offers insights into how subscription plans relate to sales performance, providing a more comprehensive view of customer engagement and product uptake. Customer Experience can use the heatmap to pinpoint states and regions where customer engagement with specific contract plans is high, helping to strategize around customer retention and satisfaction. The Sales can analyze the heatmap to identify regions with strong demand for particular plans, supporting targeted sales and promotional campaigns. The Regional VPs can use the heatmap to monitor regional performance, adjusting their operations and policies based on which contract plans perform best in their area. The Data Analytics Peers can dive deeper into the correlation between contract plan popularity, churn, and sales revenue, providing actionable recommendations for the business. This heatmap offers a geographically contextualized understanding of contract plan distribution across regions, guiding strategic decisions for customer engagement and sales initiatives.

The heatmap (Contract Plan Subscription by Region) and side-by-side bar chart (Churn and Retention by Region) provide complementary insights into the data store. The heatmap offers a

geographical view of contract plan distribution and engagement. At the same time, the bar chart highlights churn and retention rates, which are critical for driving customer engagement and sales decisions. These visuals equip executive leaders with the insights to make data-driven, region-specific decisions that align with the company's customer engagement and sales objectives.

2. Side-by-Side Bar Chart: Churn and Retention Rate by Region

This bar chart presents a comparative view of churn and retention rates across different regions, providing a straightforward way to assess which regions are performing well and which ones need attention. The tooltip displaying churn and retention rates by states within each region allows for a deeper understanding of localized trends. This visualization is essential for helping the SVP for Customer Experience identify high-risk areas, the EVP of Sales to refine customer retention strategies, and the Regional VPs to implement region-specific policies.

The SVP for Customer Experience can use the chart to understand where retention strategies must be enhanced, focusing efforts on high-churn regions. The EVP of Sales can utilize the data to tailor promotional campaigns and sales strategies to regions with high churn and low retention. The Regional VPs can identify issues affecting churn in their regions, enabling them to adapt operational tactics accordingly. The Data Analytics Peers can dive into the reasons behind churn and retention trends and recommend ways to improve customer engagement and satisfaction. This bar chart clearly highlights regional disparities in churn and retention, enabling the business to take actionable steps to improve customer engagement and sales performance.

D: SOURCES

Shacham Omer (2018) USA States to Region (States.csv)

<https://www.kaggle.com/datasets/omer2040/usa-states-to-region/data>

E: PROFESSIONAL COMMUNICATION